

Change of Axes

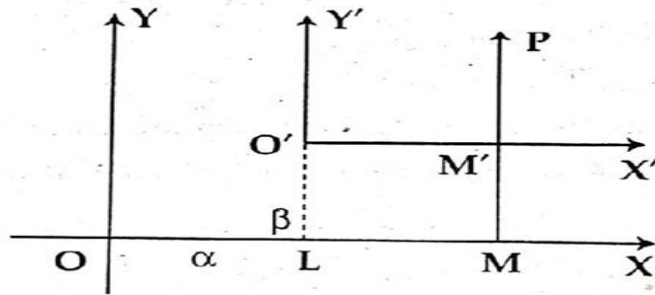
Transformation of Coordinates

The coordinates of a point or the equation of a curve is always determined with respect to a particular system of axes and a particular origin. If the system of axes is changed (which is generally done to simplify the given problem) the coordinates of the point or the equation of the curve also changes.

The system of axes can be altered in the following three ways:

1. Change of Origin (Translation of Axes)
2. Rotation of Axes (Origin fixed)
3. Change of Origin with the change of the direction of Axes

Change of Origin



Let us take a new pair of axes $O'X'$ and $O'Y'$ parallel to the old pair OX and OY ; O' being the new origin whose co-ordinates are (α, β) referred to $O'X'$ and $O'Y'$.

Let (x', y') be the co-ordinates referred to the axes $O'X'$ and $O'Y'$ of a point P , whose co-ordinates referred to the old axes are (x, y) .

It is required to transform the co-ordinates (x, y) in terms of (x', y') .

From O' and P draw $O'L$ and PM perpendiculars to OX . Let PM meet $O'X'$ in M' .

Then $OL = \alpha, LO' = \beta, OM = x, MP = y$

Also $O'M' = x'$ and $M'P = y'$

Therefore, $OM = OL + LM = OL + O'M'$

$$\therefore x = \alpha + x'$$

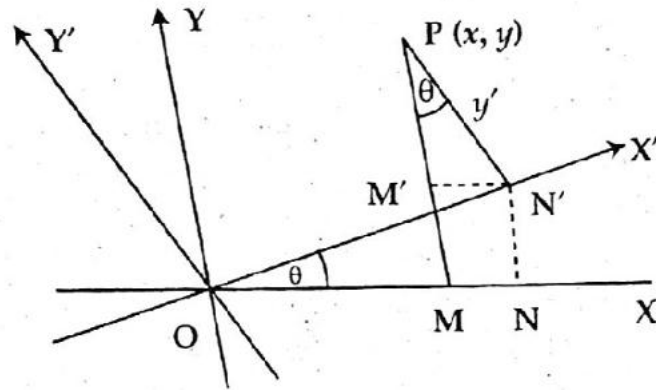
Similarly, $MP = MM' + M'P = LO' + M'P$

$$\therefore y = \beta + y'$$

The transformed co-ordinates are

$$\left. \begin{aligned} x' &= x - \alpha \\ y' &= y - \beta \end{aligned} \right\}$$

Rotation of Axes (Origin Fixed)



Let P be any point in the plane and let its co-ordinates referred to the old axes be (x, y) , and referred to the new axes be (x', y') .

Let us try to determine x and y in terms of x', y' and θ . Draw PM perpendicular to OX, PN' perpendicular to OX', and N'M' parallel to OX.

$$\begin{aligned} \text{Then } x &= OM = ON - MN = ON - M'N' \\ &= ON' \cos \theta - PN' \sin \theta = x' \cos \theta - y' \sin \theta \end{aligned}$$

$$\begin{aligned} y = MP &= MM' + M'P = NN' + M'P \\ &= ON' \sin \theta + PN' \cos \theta = x' \sin \theta + y' \cos \theta \end{aligned}$$

Hence, the formula for the rotation of axes through an angle θ are:

$$x = x' \cos \theta - y' \sin \theta$$

$$y = x' \sin \theta + y' \cos \theta$$

Change of Origin with the change of the direction of axes

If the origin shifted to the point (α, β) and then the axes are rotated through an angle θ .

The formula is transformed to:

$$x = \alpha + x' \cos \theta - y' \sin \theta$$

$$y = \beta + x' \sin \theta + y' \cos \theta$$

5. Transform the equation $11x^2 + 24xy + 4y^2 - 20x - 40y - 5 = 0$ to rectangular axes through the point $(2, -1)$ and inclined at an angle $\tan^{-1}(-\frac{4}{3})$.

Given equation, $11x^2 + 24xy + 4y^2 - 20x - 40y - 5 = 0$

For the first transformation, we have $x = x_1 + 2$; $y = y_1 - 1$

Therefore, the given equation becomes,

$$\begin{aligned} & 11(x_1+2)^2 + 24(x_1+2)(y_1-1) + 4(y_1-1)^2 - 20(x_1+2) - 40(y_1-1) - 5 = 0 \\ & 11(x_1^2 + 4x_1 + 4) + 24(x_1y_1 - x_1 + 2y_1 - 2) + 4(y_1^2 - 2y_1 + 1) - 20(x_1+2) - 40(y_1-1) - 5 = 0 \\ & 11x_1^2 + 24x_1y_1 + 4y_1^2 - 5 = 0 \end{aligned}$$

Removing subscript, we get

$$11x^2 + 24xy + 4y^2 - 5 = 0 \quad \dots \quad (1)$$

For the second transformation, we have

$$\tan\theta = -4/3, \text{ therefore, } \sin\theta = -4/5, \quad \cos\theta = 3/5$$

Now, $x = x_1 \cos\theta - y_1 \sin\theta = (3/5)x_1 + (4/5)y_1$

$$y = x_1 \sin\theta + y_1 \cos\theta = (-4/5)x_1 + (3/5)y_1$$

Using these, equation (1) becomes,

$$\begin{aligned} & 11\{(3/5)x_1 + (4/5)y_1\}^2 + 24\{(3/5)x_1 + (4/5)y_1\}\{(-4/5)x_1 + (3/5)y_1\} + 4\{(-4/5)x_1 + (3/5)y_1\}^2 - 5 = 0 \\ & 99/25x_1^2 + 176/25y_1^2 + 264/25x_1y_1 + 24\{-12/25x_1^2 + 9/25x_1y_1 - 16/25x_1y_1 + 12/25y_1^2\} + \\ & 64/25x_1^2 + 36/25y_1^2 - 96/25x_1y_1 - 5 = 0 \\ & 99x_1^2 + 176y_1^2 + 264x_1y_1 - 288x_1^2 - 168x_1y_1 + 288y_1^2 + 64x_1^2 + 36y_1^2 - 96x_1y_1 - 125 = 0 \\ & -125x_1^2 + 500y_1^2 - 125 = 0 \\ & x_1^2 - 4y_1^2 + 1 = 0 \end{aligned}$$

Removing subscript, we get

$$x^2 - 4y^2 + 1 = 0$$

For the second transformation, we have

$$\tan\theta = -4/3, \text{ therefore, } \sin\theta = 4/5, \quad \cos\theta = -3/5$$

Now, $x = x_1 \cos\theta - y_1 \sin\theta = (-3/5)x_1 - (4/5)y_1$

$$y = x_1 \sin\theta + y_1 \cos\theta = (4/5)x_1 - (3/5)y_1$$

Using these, equation (1) becomes,

$$\begin{aligned} & 11\{(-3/5)x_1 - (4/5)y_1\}^2 + 24\{(-3/5)x_1 - (4/5)y_1\}\{(4/5)x_1 - (3/5)y_1\} + 4\{(4/5)x_1 - (3/5)y_1\}^2 - 5 = 0 \\ & 99/25x_1^2 + 176/25y_1^2 + 264/25x_1y_1 + 24\{-12/25x_1^2 + 9/25x_1y_1 - 16/25x_1y_1 + 12/25y_1^2\} + \\ & 64/25x_1^2 + 36/25y_1^2 - 96/25x_1y_1 - 5 = 0 \\ & 99x_1^2 + 176y_1^2 + 264x_1y_1 - 288x_1^2 - 168x_1y_1 + 288y_1^2 + 64x_1^2 + 36y_1^2 - 96x_1y_1 - 125 = 0 \\ & -125x_1^2 + 500y_1^2 - 125 = 0 \\ & x_1^2 - 4y_1^2 + 1 = 0 \end{aligned}$$

Removing subscript, we get

$$x^2 - 4y^2 + 1 = 0$$

Exercise:

Through what angle should the axes be rotated to remove the term containing xy in $11x^2 + 4xy + 14y^2 = 5$

Let the axes be turned through an angle θ , then by putting

$$x = x' \cos \theta - y' \sin \theta$$

$$y = x' \sin \theta + y' \cos \theta$$

the given expression reduces to

$$11(x' \cos \theta - y' \sin \theta)^2 + 4(x' \cos \theta - y' \sin \theta)(x' \sin \theta + y' \cos \theta) + 14(x' \sin \theta + y' \cos \theta)^2 = 5$$

$$11(x'^2 \cos^2 \theta - 2x'y' \sin \theta \cos \theta + y'^2 \sin^2 \theta) + 4(x'^2 \cos \theta \sin \theta + x'y' \cos^2 \theta - x'y' \sin^2 \theta - y'^2 \sin \theta \cos \theta) + 14(x'^2 \sin^2 \theta + 2x'y' \sin \theta \cos \theta + y'^2 \cos^2 \theta) = 5$$

If the term xy is to be absent, then equating the coefficient of xy to zero, we have,

$$-22 \sin \theta \cos \theta + 4 \cos^2 \theta - 4 \sin^2 \theta + 28 \sin \theta \cos \theta = 0$$

$$4 \cos 2\theta + 6 \sin \theta \cos \theta = 0$$

$$4 \cos 2\theta + 3 \sin 2\theta = 0$$

$$4 + 3 \tan 2\theta = 0$$

$$\tan 2\theta = -4/3$$

$$\theta = \frac{1}{2} \tan^{-1}(-4/3), \text{ which is the required angle.}$$

Exercise:

1. Determine the equation of the curve $2x^2 + 3y^2 - 8x + 6y - 7 = 0$ when the origin is transferred to the point $(2, -1)$.

Ans: $2x^2 + 3y^2 = 18$.

2. Determine the equation of the parabola $x^2 - 2xy + y^2 + 2x - 4y + 3 = 0$ after rotating the axes through 45° .

Ans: $2y^2 - \sqrt{2}x - 3\sqrt{2}y + 3 = 0$.

3. Transform the equation $9x^2 + 24xy + 2y^2 - 6x + 20y + 41 = 0$ in rectangular coordinates so as to remove the terms in x, y and xy .

Solution:

Given,

$$9x^2 + 24xy + 2y^2 - 6x + 20y + 41 = 0 \dots\dots\dots(1)$$

Putting $x = x + \alpha$ and $y = y + \beta$ in equation (1),

$$9(x + \alpha)^2 + 24(x + \alpha)(y + \beta) + 2(y + \beta)^2 - 6(x + \alpha) + 20(y + \beta) + 41 = 0$$

$$\Rightarrow 9(x^2 + 2x\alpha + \alpha^2) + 24(xy + x\beta + y\alpha + \alpha\beta) + 2(y^2 + 2y\beta + \beta^2) - 6(x + \alpha) + 20(y + \beta) + 41 = 0$$

$$\Rightarrow 9x^2 + 24xy + 2y^2 + x(18\alpha + 24\beta - 6) + y(24\alpha + 4\beta + 20) + 9\alpha^2 + 24\alpha\beta + 2\beta^2 - 6\alpha + 20\beta + 41 = 0 \dots\dots\dots(2)$$

The terms of x and y in equation (2) will be absent if

$$18\alpha + 24\beta - 6 = 0$$

And $24\alpha + 4\beta + 20 = 0$

Solving these two equations we get,

$$\therefore \alpha = -1 \text{ and } \beta = 1$$

Putting $\alpha = -1$ and $\beta = 1$ in equation (2),

$$9x^2 + 24xy + 2y^2 + 9 - 24 + 2 + 6 + 20 + 41 = 0$$

$$\Rightarrow 9x^2 + 24xy + 2y^2 + 54 = 0 \dots\dots\dots(3)$$

Solving these two equations we get,

$$\therefore \alpha = -1 \text{ and } \beta = 1$$

Putting $\alpha = -1$ and $\beta = 1$ in equation (2),

$$9x^2 + 24xy + 2y^2 + 9 - 24 + 2 + 6 + 20 + 41 = 0$$

$$\Rightarrow 9x^2 + 24xy + 2y^2 + 54 = 0 \dots\dots\dots(3)$$

Now Putting $x = x\cos\theta - y\sin\theta$ and $y = x\sin\theta + y\cos\theta$ in equation (3),

$$9(x\cos\theta - y\sin\theta)^2 + 24(x\cos\theta - y\sin\theta)(x\sin\theta + y\cos\theta) + 2(x\sin\theta + y\cos\theta)^2 + 54 = 0$$

$$\Rightarrow 9(x^2\cos^2\theta - 2xysin\theta\cos\theta + y^2\sin^2\theta) + 24(x^2\sin\theta\cos\theta + xycos^2\theta - xysin^2\theta - y^2\sin\theta\cos\theta) + 2(x^2\sin^2\theta + 2xysin\theta\cos\theta + y^2\cos^2\theta) + 54 = 0$$

$$\Rightarrow x^2(9\cos^2\theta + 24\sin\theta\cos\theta + 2\sin^2\theta) + xy(-18\sin\theta\cos\theta + 24\cos^2\theta - 24\sin^2\theta + 4\sin\theta\cos\theta) + y^2(9\sin^2\theta - 24\sin\theta\cos\theta + 2\cos^2\theta) + 54 = 0 \dots\dots\dots(4)$$

$$\begin{aligned}
&\Rightarrow 24 \cos^2 \theta - 14 \sin \theta \cos \theta - 24 \sin^2 \theta = 0 \\
&\Rightarrow 12 \cos^2 \theta - 7 \sin \theta \cos \theta - 12 \sin^2 \theta = 0 \\
&\Rightarrow 12 \cos^2 \theta - 16 \sin \theta \cos \theta + 9 \sin \theta \cos \theta - 12 \sin^2 \theta = 0 \\
&\Rightarrow 4 \cos \theta (3 \cos \theta - 4 \sin \theta) + 3 \sin \theta (3 \cos \theta - 4 \sin \theta) = 0 \\
&\Rightarrow (3 \cos \theta - 4 \sin \theta)(4 \cos \theta + 3 \sin \theta) = 0
\end{aligned}$$

$$\therefore (4 \cos \theta + 3 \sin \theta) = 0$$

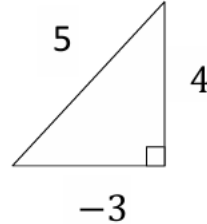
$$\Rightarrow \tan \theta = -\frac{4}{3}$$

$$\text{Or, } (3 \cos \theta - 4 \sin \theta) = 0$$

$$\Rightarrow \tan \theta = \frac{3}{4}$$

$$\text{when } \tan \theta = -\frac{4}{3}$$

$$\therefore \sin \theta = -\frac{4}{5} \text{ and } \cos \theta = \frac{3}{5}$$



Putting $\sin \theta = -\frac{4}{5}$ and $\cos \theta = \frac{3}{5}$ in equation (4),

$$\begin{aligned}
&x^2 \left(9 \cdot \frac{9}{25} + 24 \left(-\frac{4}{5} \right) \frac{3}{5} + 2 \frac{16}{25} \right) + \\
&xy \left(-18 \left(-\frac{4}{5} \right) \frac{3}{5} + 24 \cdot \frac{9}{25} - 24 \left(-\frac{4}{5} \right)^2 + \right. \\
&\left. 4 \left(-\frac{4}{5} \right) \cdot \left(\frac{3}{5} \right) \right) + y^2 \left(9 \cdot \frac{16}{25} - 24 \left(-\frac{4}{5} \right) \cdot \frac{3}{5} + \right. \\
&\left. 2 \left(\frac{3}{5} \right)^2 \right) + 54 = 0
\end{aligned}$$

$$\Rightarrow -7x^2 + 18y^2 + 54 = 0$$

$$\Rightarrow 18y^2 - 7x^2 + 54 = 0$$

This is our required transformed equation.

Invariants

Art 35. If by the rotation of the rectangular coordinates axes about the origin, the expression $ax^2 + 2hxy + by^2$ changes to $a'x'^2 + 2h'x'y' + b'y'^2$ then show that $a + b = a' + b'$ and $a - b - h^2 = a'b' - h'^2$

Let (x, y) be the coordinates of a point P referred to a set of rectangular axes. If the axes are rotated through an angle θ about the origin, let the coordinates of the same point P be (x', y') referred to the new system of rectangular axes. Then we have,

$$\begin{aligned}x &= x' \cos \theta - y' \sin \theta \\y &= x' \sin \theta + y' \cos \theta\end{aligned}$$

By using the above transformation, the expression $ax^2 + 2hxy + by^2$ becomes,

$$\begin{aligned}&a(x' \cos \theta - y' \sin \theta)^2 + 2h(x' \cos \theta - y' \sin \theta)(x' \sin \theta + y' \cos \theta) + b(x' \sin \theta + y' \cos \theta)^2 \\&= (a \cos^2 \theta + 2h \sin \theta \cos \theta + b \sin^2 \theta)x'^2 + 2x'y'(-a \sin \theta \cos \theta + h \cos^2 \theta - h \sin^2 \theta + b \sin \theta \cos \theta) \\&\quad + (a \sin^2 \theta - 2h \sin \theta \cos \theta + b \cos^2 \theta)y'^2 \\&= a'x'^2 + 2h'x'y' + b'y'^2 \\&\text{where } a' = a \cos^2 \theta + 2h \sin \theta \cos \theta + b \sin^2 \theta \\&\quad b' = a \sin^2 \theta - 2h \sin \theta \cos \theta + b \cos^2 \theta \\&\quad h' = -a \sin \theta \cos \theta + h \cos^2 \theta - h \sin^2 \theta + b \sin \theta \cos \theta\end{aligned}$$

$$\begin{aligned}\text{Now, } a' + b' &= a(\cos^2 \theta + \sin^2 \theta) + b(\sin^2 \theta + \cos^2 \theta) = a + b \\ \text{Again, } 2a' &= 2a \cos^2 \theta + 4h \sin \theta \cos \theta + 2b \sin^2 \theta \\ &= a(1 + \cos 2\theta) + 2h \sin 2\theta + b(1 - \cos 2\theta) \\ &= (a + b) + \{2h \sin 2\theta + (a - b) \cos 2\theta\}\end{aligned}$$

$$\text{Similarly, } 2b' = (a + b) - \{2h \sin 2\theta + (a - b) \cos 2\theta\}$$

$$\begin{aligned}\text{Now, } 4(a'b' - h'^2) &= (a + b)^2 - \{2h \sin 2\theta + (a - b) \cos 2\theta\}^2 - \{2h \cos 2\theta - (a - b) \sin 2\theta\}^2 \\ &= (a + b)^2 - 4h^2 - (a - b)^2 \\ &= 4ab - 4h^2\end{aligned}$$

$$\text{Or, } a'b' - h'^2 = ab - h^2$$

The two quantities $a+b$ and $ab - h^2$ for the expression $ax^2 + 2hxy + by^2$ are called invariants of transformation from one system of rectangular axes because their values remain unchanged by the transformation.

By transforming the parallel axes through a properly chosen point (h, k) prove that the equation $12x^2 - 10xy + 2y^2 + 11x - 5y + 2 = 0$ can be reduced to one containing only terms of the second degree.

Let the origin be shifted to (h,k), then referred to parallel axes through (h,k), the given equation transformed into

$$12(x'+h)^2 - 10(x'+h)(y'+k) + 2(y'+k)^2 + 11(x'+h) - 5(y'+k) + 2 = 0$$

$$\text{Or, } 12x'^2 + 2y'^2 - 10x'y' + x'(24h - 10k + 11) + y'(-10h + 4k - 5) + 12h^2 - 10hk + 2k^2 + 11h - 5k + 2 = 0 \quad \dots \dots (1)$$

If (1) is to be reduced to one containing only terms of the second degree/ If (1) is a homogeneous equation of second degree, then the co efficient of x, co efficient of y and the constant term should be separately zero. i.e.

$$24h - 10k + 11 = 0 \quad \dots \dots (2)$$

$$-10h + 4k - 5 = 0 \quad \dots \dots (3)$$

$$12h^2 - 10hk + 2k^2 + 11h - 5k + 2 = 0 \quad \dots \dots (4)$$

Solving (2) and (3)

$$h = -3/2 ; \quad k = -5/2$$

Substituting $h = -3/2 ; \quad k = -5/2$ in L.H.S. of (4)

L.H.S. =

$$= 0$$

i.e. the value of h and k satisfies equation (4).

Hence, the given equation can be reduced to one containing only terms of the second degree.

Find the angle through which the axes must be turned so that the expression $ax^2 + 2hxy + by^2$ become an expression in which the term xy is absent.

Let the axes be turned through an angle θ , then by putting

$$x = x' \cos \theta - y' \sin \theta$$

$$y = x' \sin \theta + y' \cos \theta$$

the given expression reduces to

$$\begin{aligned} & a(x' \cos \theta - y' \sin \theta)^2 + 2h(x' \cos \theta - y' \sin \theta)(x' \sin \theta + y' \cos \theta) + b(x' \sin \theta + y' \cos \theta)^2 \\ &= (a \cos^2 \theta + 2h \sin \theta \cos \theta + b \sin^2 \theta)x'^2 + 2x'y'(-a \sin \theta \cos \theta + h \cos^2 \theta - h \sin^2 \theta + b \sin \theta \cos \theta) \\ &+ (a \sin^2 \theta - 2h \sin \theta \cos \theta + b \cos^2 \theta)y'^2 \end{aligned}$$

If the term xy is to be absent, then equating the co efficient of xy to zero, we have,

$$-a \sin \theta \cos \theta + h \cos^2 \theta - h \sin^2 \theta + b \sin \theta \cos \theta = 0$$

$$\text{Or, } h \cos 2\theta = \left(\frac{1}{2}\right)(a - b) \sin 2\theta$$

$$\text{Or, } \tan 2\theta = \{2h / (a-b)\}$$

$$\text{Or, } \theta = (1/2) \tan^{-1} \{2h / (a-b)\}, \text{ which is the required angle.}$$

Exercise 1: Determine the angle through which the axes must be rotated to remove the xy term in equation $7x^2 - 6\sqrt{3}xy + 13y^2 = 16$.

Exercise 2: Determine the equation of the curve $4xy - 3x^2 = 1$ after rotation of axes through $\sin^{-1}(-\frac{2}{\sqrt{5}})$.

Exercise 3: Determine the equation of the curve $9x^2 + 24xy + 2y^2 + 54 = 0$, when the origin is transferred to the point $(-1, 1)$, where the direction of axes remains unaltered.

Ans: $9x^2 + 24xy + 2y^2 + 54 = 0$

Exercise 4: If the axes are turned through an angle $\tan^{-1} 2$, then what does the equation $4xy - 3x^2 = 4$ become?

Ans: $x^2 - 4y^2 - 4 = 0$

Exercise 5: Transform the equation $17x^2 + 18xy - 7y^2 - 16x - 32y - 18 = 0$ to one in which there is no term involving x, y and xy .

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